Portfolio Optimised with Genetic Algorithms

Course: Optimisation Methods

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> English translation of the original Spanish coursework report. File and function names refer to the original 2006 MATLAB code, with their meaning given in brackets. See the repository README for how they map onto the refactored code.

Files

The implementation consists of the following files. Their purpose and contents are explained below.

Data generation files

`GeneracionDatosBolsaMadrid.m` [Madrid Stock Exchange data generation] and `GeneracionDatosPort.m` [port data generation] generate the covariance-matrix and mean-return files for each context.

Data files

The `datos` [data] folder holds one folder per context (`bolsaMadrid` [Madrid Stock Exchange], `port1`, `port2`, `port3`, `port4`). Each of these folders contains the mean-return, correlation and covariance files.

The solver functions work on whichever context they are given. Besides creating a file with the target return, they copy that context's correlation matrix and mean-return vector into the `./datos/actual` [data/current] folder (`RendimientoObjetivo.txt` [target return], `MatrizCovarianzas.txt` [covariance matrix], `RendimientosMedios.txt` [mean returns]).

Genetic algorithm function files

Our genetic algorithm is implemented with the MATLAB `ga` command. Its inputs are the function to minimise and a structure holding all the configuration parameters, including the functions that create and modify the population.

`CrearPoblacionInicial.m` [create initial population]

The function has the following signature:

```
function Population = CrearPoblacionInicial(GenomeLength, FitnessFcn, options)
%Genomelength   -- Number of independent variables for the fitness function
%FitnessFcn     -- Fitness function options
%options        -- Options structure
```

where:

- `GenomeLength` is the cardinality of the problem: the number of assets to invest in.
- `FitnessFcn` is the risk function. The lower its value, the higher the individual's score.

`CruzarPadres.m` [cross parents]

Performs an R3 crossover. It avoids pairing two identical parents for as long as possible, that is, until only parents with the same genome remain.

`GenerarMutacion.m` [generate mutation]

A mutation inserts into the genome an asset that was not already in it, removing another one.

`Riesgo.m` [risk]

The function minimised by the genetic algorithm. It reads the target return, covariance matrix and mean returns from the `./datos/actual` folder. It returns only the function value, not the weight vector, to match the signature the `ga` command expects.

`Riesgo_con_Devolucion_de_pesos.m` [risk, also returning the weights]

Same as the previous function, but it also returns the weight vector.

Solver files

These automate solving one of the five problems (`bolsaMadrid`, `porti`) for a given cardinality, with either the quadratic solver or the genetic solver. The quadratic solver is only feasible for the first problem.

- `Resolucion_Cuadratica_Minimo_Riesgo.m` [quadratic solution, minimum risk]
- `Resolucion_Genetica_Minimo_Riesgo.m` [genetic solution, minimum risk]

Chart generation and display files

`GenerarGrafica.m` [generate chart]	For one problem and one cardinality.
`GenerarTodaslasGraficas.m` [generate all charts]	For every problem and all of its cardinalities.
`PresentarGrafica.m` [show chart]	For one problem and one cardinality.
`Presentar_Todas_Graficas.m` [show all charts]	For the port problems with cardinality 3 to 9.

Algorithm details

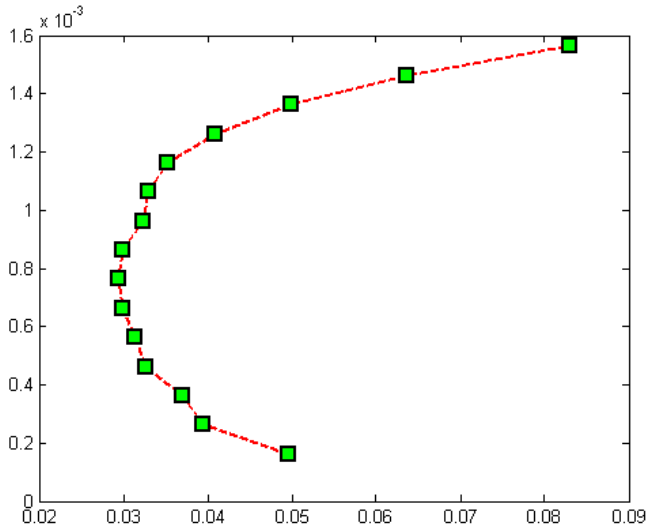
The algorithm is a genetic algorithm built on the MATLAB `ga` command. The options structure used is:

```
%   PopulationType: 'custom'
%   PopInitRange: [minimum values; maximum values]
%   PopulationSize: 2 times the number of available assets
%   EliteCount: 2
%   CrossoverFraction: 0.8000
%   MigrationDirection: 'forward'
%   MigrationInterval: 20
%   MigrationFraction: 0.2000
%   Generations: 10
%   TimeLimit: Inf
```

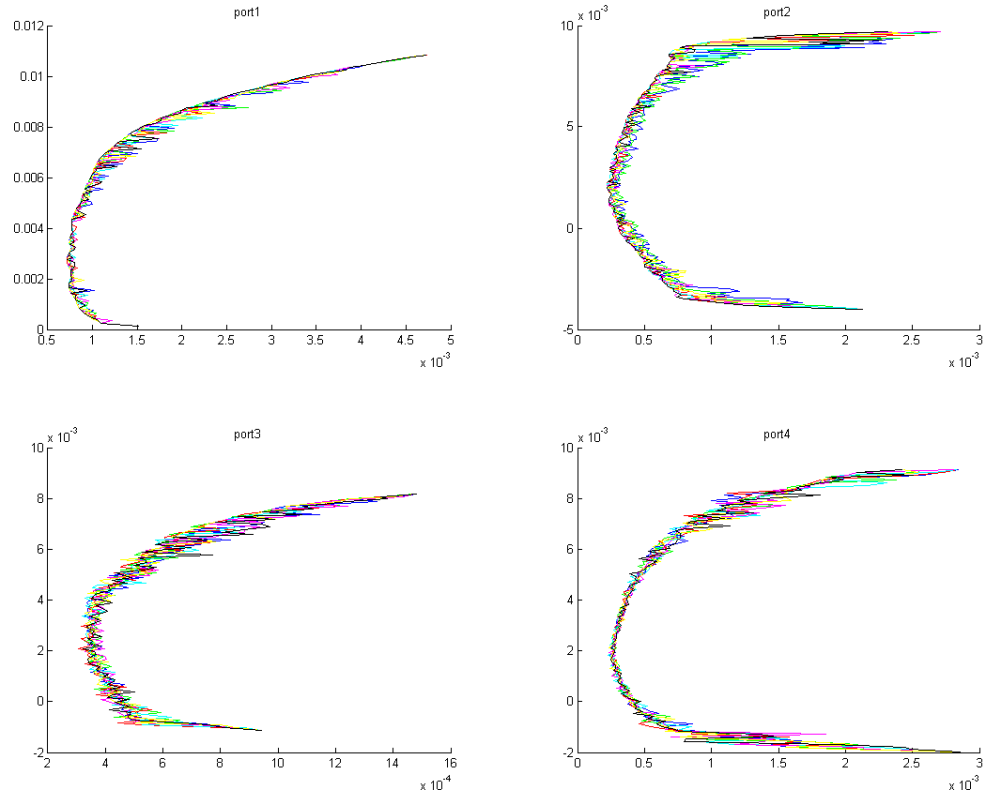
```
%   FitnessLimit: -Inf
%   StallLimitG: 5
%   StallLimitS: 20
%   InitialPopulation: []
%   InitialScores: []
%   PlotInterval: 1
%   CreationFcn: @CrearPoblacionInicial
%   FitnessScalingFcn: @fitscalingrank
%   SelectionFcn: @selectionstochunif
%   CrossoverFcn: @CruzarPadres
%   MutationFcn:  @GenerarMutacion
%   HybridFcn: []
%   Display: 'final'
%   PlotFcns: []
%   OutputFcns: []
%   Vectorized: 'off'
```

Charts

The following chart shows the result for the 5 Madrid Stock Exchange stocks with cardinality 3 (`./datos/bolsaMadrid/coordenadas_rend_riesgo_cardinalidad3.txt` [return–risk coordinates, cardinality 3]). The horizontal axis is risk (variance) and the vertical axis is mean return.



The following charts show the results for the problems port*i* (*i* = 1..4) with cardinality 3 to 9 (3 blue, 4 green, 5 red, 6 cyan, 7 magenta, 8 yellow). The data is in the files `./datos/porti/coordenadas_rend_riesgo_cardinalidad`*n* (*i* = 1..4, *n* = 3..9).



Comments

The results can be improved. The first step would be a new crossover function: the current implementation is R3 crossover, which RAR crossover would clearly improve on.

Another route to improvement is elite children. These would come from the solution of the same problem for a target return very close to the current one.